Analyzing Narrative Structures in Text-based Game Walkthroughs through a Computational Approach

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Case study and research questions

Walkthroughs produced by game players have become an important form of paratext that documents gameplay, strategies for completing digital games, and the storyline. Although not authored by a game designer, walkthroughs reflect the passion of players about how they interpret a game's structure, record essential narrative moments, and offer suggestions for other players (Newman 2011). Particularly in text-based games, where narrative plots and choice structures are primarily represented by written description, walkthroughs serve as the summary content that players consider to have narrative meaning.

The Game Walkthrough Corpus (GWTC), developed by Tiepmar and colleagues, provides a framework for processing walkthroughs as a research dataset and highlights the value of examining walkthrough language to understand game discourse (Burghard and Tiepmar 2021). Research in web-based corpora shows the methodological challenges of working with dynamic, user-generated texts, including structural inconsistency and variation in authorial style (Krause 2016). Building on these methodological insights, this project investigates how walkthroughs for text-based games reflect differences in narrative structures between descriptive story segments and gameplay decision-making segments.

This methodology incorporates computational tools, such as text extraction, linguistic preprocessing, sentiment analysis, and topic modelling, alongside qualitative corpus reading. The hybrid approach aligns with corpus work in digital humanities and textual research, and is also suited for providing standardized, machine-readable representations of complex textual

objects (Tiepmar 2014). This focused research question allows a small but meaningful dataset to be processed within 2-4 weeks, following GWTC's recommendations for manageable digital corpora (Burghard and Tiepmar 2021).

The methodological design draws directly from the practices used in constructing the GWTC (Burghard and Tiepmar 2021), corpus linguistics literature (Krause 2016).

Description of data

The dataset consists of 6-8 walkthroughs for English-language text-based or text-heavy narrative games. The sources include:

- Entries from the Game Walkthrough Corpus (GWTC)
- Publicly accessible walkthroughs from GameFAQs, Fandom wikis, and independent blogs

Walkthroughs are selected based on:

- Text-heavy gameplay where narrative is central
- Presence of both narrative descriptions and choice-related instructions
- Sufficient length of linguistic analysis (4,000-7,000 words each)
- Public accessibility and compliance with ethical use

Estimated total corpus size: 30,000-50,000 words.

Each document is stored with metadata, including URL, author (if known), access date, section structure, and estimated word count. Walkthroughs are considered as archival records of gameplay, documenting how players experience and summarize game narratives (Newman 2011).

Data collection process

The data collection process follows established practices in digital humanities corpus construction and replicates elements of the GWTC workflow to ensure consistency and reproducibility (Burghard and Tiepmar 2021). Walkthroughs are sourced from platforms that explicitly allow non-commercial academic reuse, and each webpage is reviewed manually to confirm that the collected text represents original player-generated content rather than copyrighted material such as game scripts.

Python's requests and BeautifulSoup libraries are used to automate retrieval, and the scraping complies with each platform's guidelines. After extraction, the raw HTML files are archived with timestamps and URLs to maintain clear provenance, following recommendations for sustainable data in web-based corpus research (Krause 2016).

In addition to searching the main walkthrough text, contextual information like user comments, update logs is also recorded when relevant, as they may influence the interpretation of narrative or structural features. Formatting elements that meaningfully contribute to gameplay structure, including numbered steps, choice trees, and section titles, are preserved. Any interactive elements that can hardly be captured through HTML scraping, such as embedding videos, are documented but omitted from the corpus.

Python application

Data cleaning and normalisation

Following GWTC's pipeline (Burghard and Tiempar 2021):

- Remove navigation elements, menus, comments, ads, and the website template.
- Strip HTML tags and convert files to UTF-8 encoding.
- Normalize whitespace, line breaks, and punctuation.
- Create two versions of each text: a raw corpus for close reading, a clean corpus for computational processing.

spaCy is used for tokenization, lemmatization, and sentence segmentation. This process aligns with methodologies for preparing web-based corpora, which emphasize cleaning inconsistent or chaotic user-generated text (Krause 2016).

Based on existing descriptions of gaming discourse that mixed narrative and instructional text, the corpus is manually and computationally annotated into two main categories:

1. Narrative Description:

    - Summaries of story events
    - Environmental description
    - Character actions and motivations
    - Emotionally relevant cues

2. Gameplay instruction:

    - Imperative forms (e.g., “choose Option A,” “go left”)
    - Puzzle explanations
    - Optimal path directions
    - Branching structure indicators

Several approaches are applied in the classification (Burghard and Tiempar 2021):

- Rule-based keyword detection
- POS-based detection and extraction (imperative verbs, modal verbs)
- Manual validation procedures

Computational Analysis Framework

Using pandas, spaCy, and NLTK to calculate the following features:

- Word frequencies
- Lexical diversity
- Average sentence length

- TF-IDF weighting (Salton and Buckley 1988)
- Distribution of imperative structures and descriptive structures

This enables direct comparison between narrative and instructional segments.

Sentiment is analyzed using:

- VADER, effective for social and informal text.

Previous scholarship indicates that walkthroughs embed both description and evaluation (Newman 2011), making sentiment a useful indicator for distinguishing narrative segments.

Expected finding:

Narrative segments exhibit higher emotional fluctuations; instructions remain largely neutral.

LDA (Latent Dirichlet Allocation) topic modeling (Blei et al. 2003) was applied using the Genism (Řehůřek 2010) implementation to identify recurring thematic clusters within the game's text description, similar to GWTC procedures (Burghard and Tiempar 2021). Topics are extracted separately for each category.

Expected topic clusters:

- Narrative: characters, events, setting
- Instruction: steps, options, tasks, mechanics

Structural mapping of walkthrough flow

1. Proportion of narrative and instructional content per section:

    - Using pandas for segmentation and matplotlib for plotting to visualize how these categories alternate across the text in a stacked bar chart.

2. Distribution of choices and decision points:

- Using regex to extract decision-related expressions (e.g., “option”, “choose”). Using pandas and matplotlib to visualize their frequency per segment in a time-series line chart, representing where branching moments cluster.

3. Emotional peaks over walkthrough progression:

- Sentiment scores are calculated with VADER, and results are visualized by matplotlib as an emotion curve, highlighting narrative or gameplay peaks.

These visual layers create a concise structural map of the walkthrough, reflecting the game's organizational structure.

Limitations

Despite the strengths of the approach, several methodological limitations must be acknowledged. Walkthroughs are user-generated documents, and their quality varies across platforms and authors. As Newman (2011) emphasizes, walkthroughs reflect individual players’ comprehension and stylistic choices. For example, the narrative density, or emotional peaks, may partly reflect an author’s writing habits rather than the game's underlying design. There are still differences in detail in walkthroughs and gameplay structure.

The distinction between “narrative” and “instructional” content can be ambiguous in practice. Some segments blend storytelling with commands, and the classification based on keywords in Python may mislabel mixed cases. Using regex patterns to detect and capture the decision markers may overlook implicit choices or complex branching structures that are not explicitly represented in the text.

The VADER model is widely used for social media text, and it was not specifically designed for gaming discourse, which has constraints on sentiment analysis on game walkthroughs. The gaming discourse often contains specialized terminology to describe the emotion, and emotional peaks observed in the walkthrough by VADER may therefore be driven by linguistic features rather than narrative or gameplay intensity.

Gaming is a dynamic, spatial, and audiovisual experience, and walkthroughs reduce part of the experience into a single written text, in digital textual representations; such a reduction simplifies complex object structure (Tiepmar 2014). The structural mapping relies on linear text analysis, which can not fully capture interactive elements of gameplay.

Expected results

Based on the combined computational and qualitative methodology, the project is expected to reveal clear differences between narrative and instructional segments within text-based game walkthroughs. It is anticipated that narrative parts will display greater lexical diversity, longer sentence structures, and more frequent use of descriptive vocabulary, reflecting their function in storytelling and constructing emotional meaning. Sentiment analysis conducted with VADER is expected to show more fluctuating emotional intensity in these narrative segments, producing identifiable emotional peaks that align with moments in the storyline (Newman 2011).

In contrast, instructional segments are expected to exhibit more uniform linguistic patterns, including higher frequencies of imperative verb forms, spatial orientation terms. TF-IDF weighting should highlight task-related terminology as central to this category (Salton and Buckley 1988). Decision-related expressions extracted through regex are likely to cluster around sections of branching gameplay, supporting the view that even general strategy texts encode the fun structure of a game (Burghard and Tiepmar 2021).

Topic modelling using LDA is expected to produce thematic clusters that are with two different text types: narrative sections will likely group around themes of characters, setting, and major events, while instruction sections will cluster around tasks, routes, steps, and mechanical explanations. The structural mapping visualizations are expected to show alternating patterns of narrative and instruction.

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